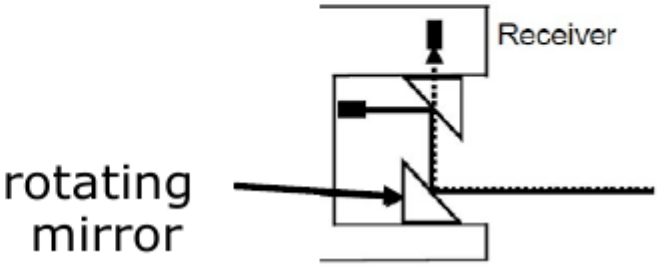
Ro06: Basic Observation Models

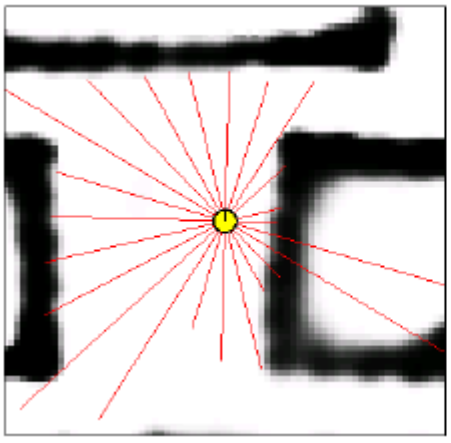
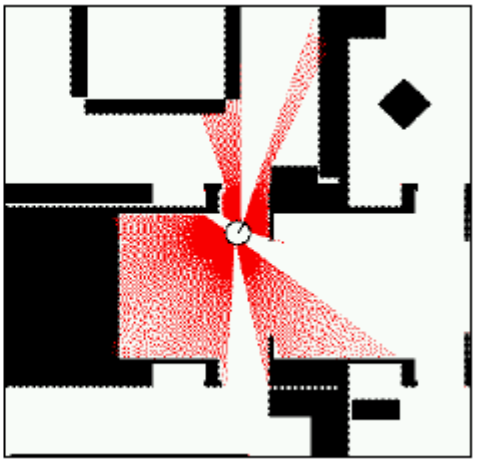
Range Sensors

Laser Range Sensors

- $d = \frac{ct}{2}$
-

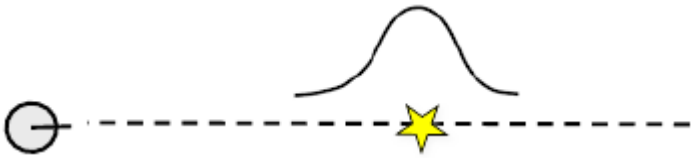
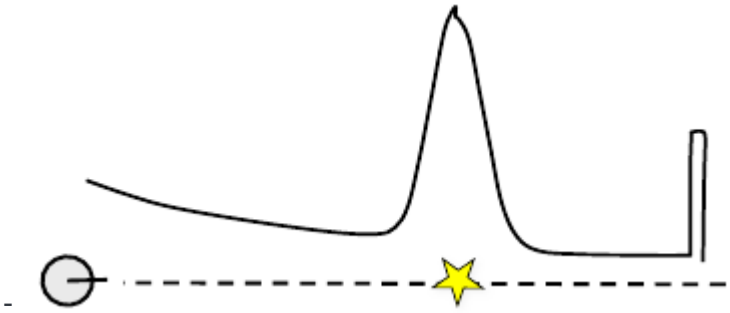
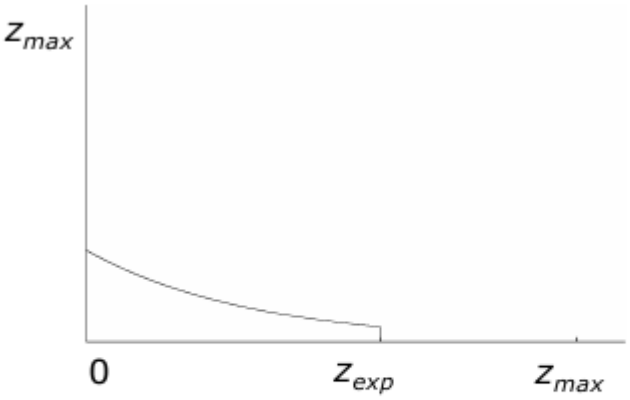


Beam-based Sensor Model

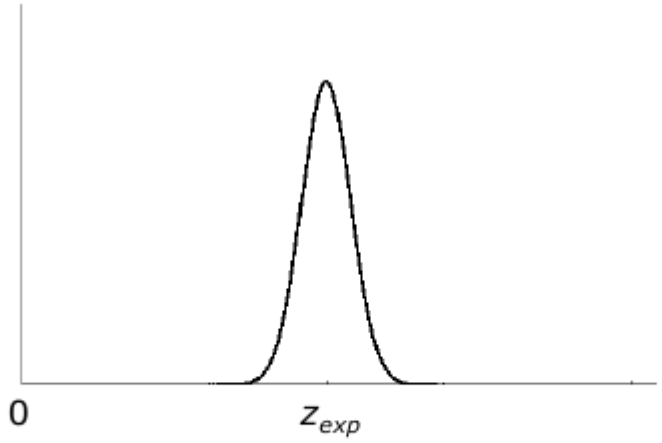
- Scan z: K measurements:
  - $z_t = \{z_t^1, ..., z_t^k\}$
- Individual measurements independent of sensor position
  - $p(z_t|x_t, m) = \prod_{i=1}^{k} p(z_t^i|x_t, m)$
- 


Ray-Cast Model

- **Simple** Ray-cast models consider **first obstacle** along LoS
  - Gaussian noise in the distance

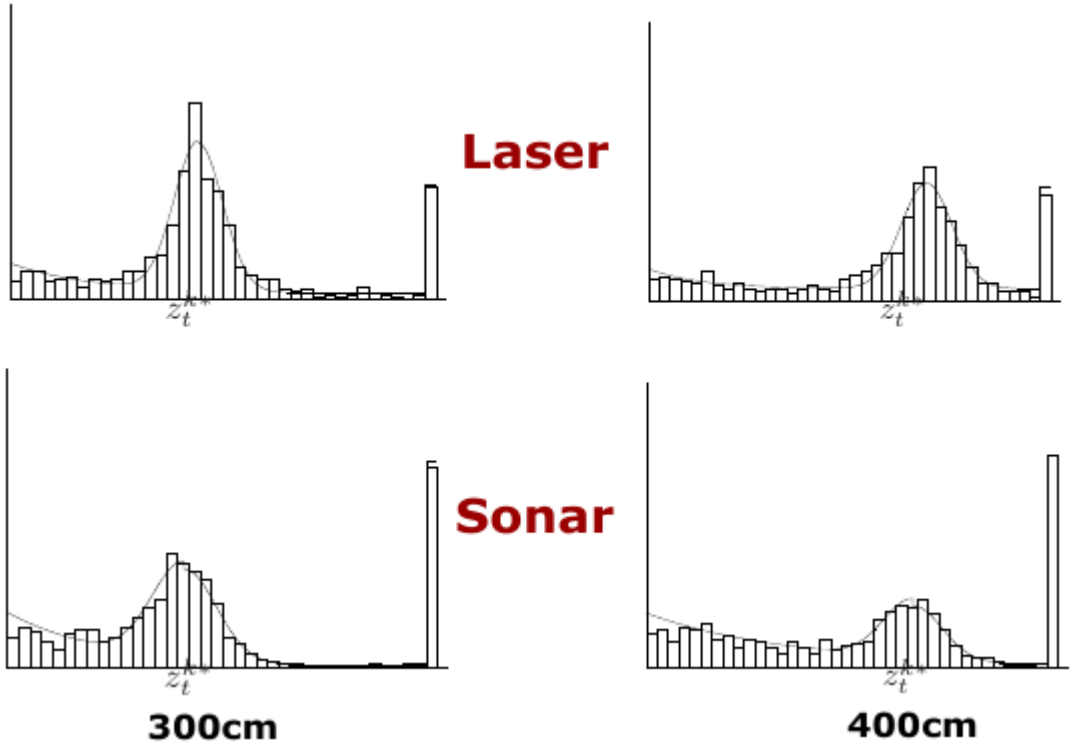
- 
- **More Advanced** Ray-Cast Model:
  - **Mixture** of 4 models
    - 
  - **Dynamic objects**
    - Account for closest obstacle in a uniform distribution of objects: Logarithmic
    - Given obstacle find bit strings from random bit string
      - 1*** -> 8
      - 01** -> 4
      - 001* -> 2
      - 0001 -> 1
      - 

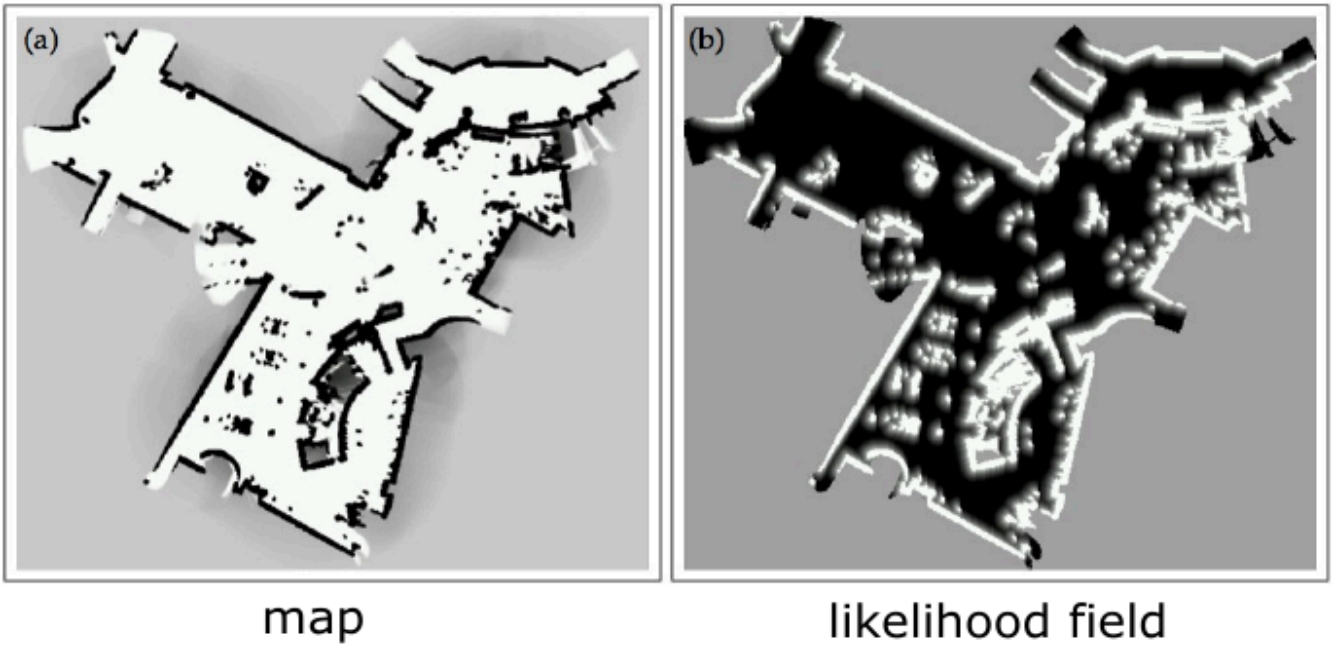
      - $$P_{unexp}(z \mid x, m) = \begin{cases} \lambda \exp(-\lambda z) & z < z_{exp} \\ 0 & \text{otherwise} \end{cases}$$
  - **Random**
    - constant
  - **Max-range**
    - constant peak at z_{max}
  - **Noise**



- $$P_{hit}(z \mid x, m) = (2\pi\sigma^2)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}\frac{(z - z_{exp})^2}{\sigma^2}\right)$$

- Similar Results for Lasers and Sonars

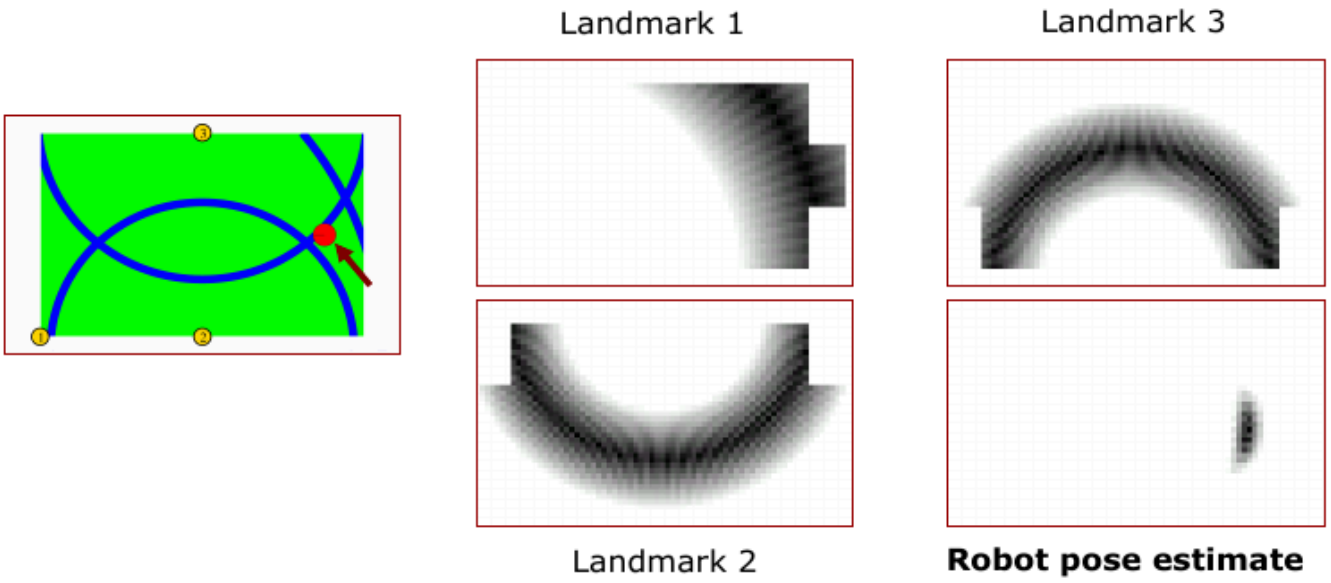


- 

Models for Landmark Observation

- Stereo Cameras Measure Point in Space via Triangulation
  - Usage of 2 cameras

Simple Range Bearing Sensors

- Model for perceiving Landmarks in 2D from
  - Pose (x, y, θ)
- Expected observation of feature j at location $(m_{j,x}, m_{j,y})^T$
  - $z_t^i = \begin{bmatrix} r_t^i \\ \phi_t^i \end{bmatrix} = \begin{bmatrix} \sqrt{(m_{j,x} - x)^2 + (m_{j,y} - y)^2} \\ atan2(m_{j,y} - y, m_{j,x} - x) - \theta \end{bmatrix}$
  - Gaussian noise with covariance Q_t in range and bearing measurement
- 

Bearing-Only sensors

- Bearing $z_t^i = \phi_t^i$

- $\phi_t^i = atan2(m_{j,y} - y, m_{j,x} - x) - \theta$

Mapping of 3D Point to Pixel Coordinate

$$\underbrace{\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}}_{\text{pixel coordinate}} = \underset{\text{transformation}}{P} \underbrace{\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}}_{\text{world coordinate}}$$

Where $P = KR[\boldsymbol{I}_3 | -\boldsymbol{X}_O]$

$$\underbrace{\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}}_{\text{2D pixel coordinate}} = \lambda \underbrace{\begin{bmatrix} c & cs & x_H \\ 0 & c(1+m) & y_H \\ 0 & 0 & 1 \end{bmatrix}}_{\text{camera calibration matrix}} \underbrace{\begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}}_{\text{rotation matrix}} \underbrace{\begin{bmatrix} 1 & 0 & 0 & -X_O \\ 0 & 1 & 0 & -Y_O \\ 0 & 0 & 1 & -Z_O \end{bmatrix}}_{\text{projection matrix and translation}} \underbrace{\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}}_{\text{3D world coordinate}}$$

- **Intrinsics** (Calibration Matrix K):
  - contains 5 paramters:
    - Camera constant c
    - Principal point: x_H, y_H
    - Scale difference m
    - shear s
- **Extrinsics** (Rotation and Translation)
  - camera-external parameters -> Position
  - Position of camera projection center $\boldsymbol{X}_O$
  - Heading of the camer R

DLT: Direct Linear Transform

- Homogeneous Projection Matrix
  - $P = KR[\boldsymbol{I}_3 | -\boldsymbol{X}_O]$
- Contains **11 parameters**
  - 6 **extrinsic parameters**: $R, \boldsymbol{X}_O$
  - 5 **intrinsic parameters**: c, x_H, y_H, m, s
- Mapping form 3D point to 2D pixel coordinate: $\boldsymbol{x} = p\boldsymbol{X}$
- **Inverse Mapping**
  - λ: factor resulting from H.C.
  - $\boldsymbol{X} = \boldsymbol{X}_O + \lambda(KR)^{-1}\boldsymbol{x}$

    - Describes line in 3D world
    - $\lambda(KR)^{-1}\boldsymbol{x}$ describes direction of ray from Camera origin $\boldsymbol{X}_O$ to 3D poing $\boldsymbol{X}$
    - Measuring directions

Summary

- Basic sensing with laser range finders and cameras
- Probabilistic observation models for such sensors
- Observataion (and motion) models are central models in the Bayes filter that need to be specified
- Raycasting models for range sensors
- Beam-endpoint models for range sensors
- Simple models for feature/point observations:
    - triangulation
    - range-only
    - bearing-only